SCHOOL OF AGRICULTURAL AND BIOSYSTEMS ENGINEERING

DEPARTMENT OF AGRICULTURAL AND BIOSYSTEMS ENGINEERING

SMART WEED DETECTION

A Project Proposal Submitted to the Department of Agricultural and Biosystems Engineering

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This project proposal report was submitted to the Department of Agricultural and Biosystems Engineering in partial fulfillment of the requirements for the award of the degree of Bachelor of Science in Agricultural and Biosystems Engineering at Jomo Kenyatta University of Agriculture and Technology.

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DECLARATION OF AUTHORSHIP

Student Confirmation:

We, Amos Kibet and Paul Kimanga M declare that this report is our original work to the best of our knowledge and has not been submitted for any degree award in any university or institution.

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Supervisor Confirmation:

This project report has been submitted for examination with my approval/knowledge as University supervisor.

Signed __________________________ Date __________________________

Prof. Ajwang

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ABSTRACT

Agriculture remains a critical pillar of Kenya's economy, contributing significantly to GDP, employment, and food security. However, rising production costs, environmental degradation, and labor shortages continue to threaten productivity. One of the most persistent challenges is weed management, which if poorly controlled reduces yields by up to 40%. Traditional methods such as manual weeding and blanket herbicide spraying are increasingly unsustainable due to high labor costs, chemical wastage, environmental pollution, and herbicide-resistant weed populations.

This project proposes the design, development, and implementation of a Smart Weed Detection system integrating Artificial Intelligence (AI) with the Internet of Things (IoT). The system employs a deep learning object detection model designed to differentiate weeds from crops based on morphological characteristics such as shape, texture, and color patterns. The AI model will be trained using high-resolution images (minimum 12 megapixels) of crops and weeds captured from local farms to ensure detailed feature extraction.

The hardware architecture centers on an ESP-32 microcontroller. The system processes continuous video feed at 30 frames per second. When a weed is detected with confidence exceeding 85%, the ESP-32 activates an LED indicator that blinks for 2 seconds, marking each detection event and providing clear validation of system performance.

The system uses breadboard, jumper wires, ESP-32, resistors, and LED indicators. AI training is conducted using Google Colab or PC with GPU. Total budget: KES 10,150. Expected outcome: a functional prototype achieving minimum 85% mean Average Precision (mAP), demonstrating viability of low-cost AI-powered precision agriculture for Kenyan smallholder farmers.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
CNN	Convolutional Neural Network
COCO	Common Objects in Context
CV	Computer Vision
FPS	Frames Per Second
GDP	Gross Domestic Product
GPIO	General Purpose Input/Output
GPU	Graphics Processing Unit
IoT	Internet of Things
KES	Kenyan Shilling
mAP	Mean Average Precision
NIR	Near-Infrared
PA	Precision Agriculture
PCB	Printed Circuit Board
RGB	Red, Green, Blue
SDGs	Sustainable Development Goals
SSWM	Site-Specific Weed Management
UART	Universal Asynchronous Receiver-Transmitter
VRT	Variable Rate Technology
YOLO	You Only Look Once
Industry 4.0	Fourth Industrial Revolution

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

Agriculture remains the cornerstone of Kenya's economy, accounting for approximately 33% of the Gross Domestic Product (GDP) and providing livelihoods for over 75% of the rural population. The sector is critical for food security, employment, and export earnings. However, agriculture faces mounting pressure to increase productivity for a growing population projected to reach 95 million by 2050, while simultaneously reducing environmental impact.

Traditional farming practices are increasingly resource-intensive and economically inefficient. Rising input costs for labor, fertilizers, and agrochemicals, combined with unpredictable weather and declining soil fertility, threaten agricultural productivity. Weed management is one of the most persistent challenges—weeds compete with crops for nutrients, water, and sunlight, causing yield reductions of 30–40% if uncontrolled. Manual weeding with hoes is labor-intensive, covering only 0.1–0.2 hectares per day per worker, with labor costs rising from KES 300–400/day in 2015 to KES 500–800/day in 2025.

Chemical control using herbicides revolutionized agriculture but relies on blanket spraying—applying chemicals uniformly across entire fields regardless of weed distribution. Research shows that weed coverage is typically patchy, occupying only 25–40% of land area. Consequently, 60–70% of herbicides are wasted on bare soil or crops, representing 15–25% of total production costs. Beyond economic waste, blanket herbicide application causes soil contamination, groundwater pollution, destruction of beneficial soil organisms, and development of herbicide-resistant weeds.

The integration of Computer Vision (CV), Artificial Intelligence (AI), and the Internet of Things (IoT) offers transformative solutions through Precision Agriculture (PA). This project develops a low-cost, locally-fabricated smart weed detection system using deep learning object detection to perform real-time weed identification, with LED indicators providing validation before scaling to automated applications.

1.2 Problem Statement

Current weed control methods suffer from critical limitations that undermine agricultural productivity and sustainability:

Economic Inefficiency: Blanket spraying wastes 60–70% of chemicals on weed-free areas. For a 2-hectare maize farm, herbicide costs range from KES 8,000–15,000 per season, with KES 5,000–10,000 wasted. Manual weeding costs KES 7,500–16,000 per cycle (15–20 person-days), with 2–3 cycles per season required.

Environmental and Health Risks: Chemical overuse causes soil degradation, water contamination, and health risks to farm workers. Herbicide residues persist in agricultural soils for 6–18 months. Herbicide-resistant weed populations are evolving, including Amaranthus hybridus and Eleusine indica.

Lack of Affordable Smart Solutions: Commercial agricultural robots cost USD 50,000–300,000 (KES 6.5–39 million), viable only for large farms exceeding 500 hectares (less than 5% of Kenyan holdings). Existing systems are designed for temperate crops and flat terrain, not African mixed cropping systems.

This project bridges this gap by creating a low-cost, AI-driven weed detection system for Kenyan smallholder conditions using affordable electronics, open-source software, and cloud-based AI training.

1.3 Objectives

1.3.1 Main Objective

To design, develop, and evaluate a Smart Weed Detection system with automated LED indicator response using Artificial Intelligence and Internet of Things technology for precision agriculture applications.

1.3.2 Specific Objectives

1. Develop and train a deep learning object detection model capable of identifying weeds versus crops with a minimum of 85% mean Average Precision (mAP) using high-resolution field images.
2. Design and implement an IoT-based control circuit using the ESP-32 microcontroller to activate an LED indicator for 2 seconds upon weed detection.
3. Evaluate system performance in terms of detection accuracy, response latency, and reliability under controlled testing conditions.

1.4 Hypothesis

The integration of a YOLO-NAS-based deep learning model with an ESP-32 IoT microcontroller can achieve a minimum detection accuracy of 85% mean Average Precision (mAP) for weed identification in castor bean fields, enabling reliable real-time LED actuation within 500 milliseconds, thereby demonstrating the feasibility of low-cost AI-powered precision weed management for Kenyan smallholder farmers.

Null Hypothesis (H0): There is no significant difference in detection accuracy between the proposed YOLO-NAS smart weed detection system and random classification, meaning the system cannot reliably distinguish weeds from crops.

Alternative Hypothesis (H1): The YOLO-NAS-based smart weed detection system achieves a statistically significant improvement in detection accuracy over random classification, with a minimum mAP of 85%, confirming that AI-driven image analysis can reliably identify weeds in field conditions.

1.5 Justification of the Study

This project aligns with UN Sustainable Development Goals: SDG 2 (Zero Hunger) by reducing crop losses, SDG 12 (Responsible Consumption) through efficient resource use, and SDG 15 (Life on Land) by protecting soil health.

Economic Impact: The system saves farmers KES 5,000–10,000 per hectare per season on herbicides (60–70% reduction). Annual savings of KES 20,000–40,000 are projected for a 2-hectare farm. Effective weed control prevents yield losses worth KES 30,000–60,000 per hectare.

Technological Innovation: The project demonstrates practical Edge AI on low-cost hardware, creates the first comprehensive annotated dataset of Kenyan crops and weeds, and establishes an IoT integration framework for agricultural automation in developing countries.

Environmental Sustainability: Reduced chemical application protects soil microorganisms, preserves biodiversity including pollinators, and manages herbicide resistance.

1.6 Scope

Target Crops and Weeds: The primary crop is Castor Bean (Ricinus communis), characterized by large star-shaped palmate leaves and reddish-purple stems. Target weeds include Amaranthus viridis/hybridus (Pigweed), Silybum marianum (Milk Thistle), Raphanus raphanistrum (Wild Radish), Parthenium hysterophorus (Congress Weed), and Trianthema portulacastrum (Horse Purslane).

Image Data Source: High-resolution images (minimum 12 megapixels) were collected from a castor bean farm in Kiambu County during daylight hours (8AM–5PM) at various crop growth stages. Data collection was carried out directly in the field, capturing real growing conditions, soil backgrounds, and natural lighting variations present on the farm. Images will be manually annotated using LabelImg software. The dataset follows a standard train/val/test split of 70%/20%/10%.

System Configuration: RGB color images processed through the YOLO-NAS object detection model analyzing plant shape, leaf texture, color patterns, and spatial arrangement to differentiate 6 classes (1 crop + 5 weed species). Hardware comprises a breadboard assembly with ESP-32 microcontroller (KES 1,400), resistors (220–330Ω), and 5mm LEDs. Total budget: KES 10,150.

Limitations: The current scope focuses on detection validation using LED indicators. It does not include mechanical spraying or physical weeding, and the model is optimized for daylight conditions and specifically trained for the castor bean crop system.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter reviews theoretical foundations, technological developments, and empirical research relevant to smart weed detection systems, covering precision agriculture evolution, deep learning for computer vision, object detection architectures, and hardware platforms for agricultural IoT applications.

2.2 Precision Agriculture and Site-Specific Weed Management

Precision Agriculture (PA) emerged in the 1980s with GPS, GIS, and variable rate equipment. It manages spatial and temporal variability within fields to optimize returns while minimizing environmental impact—applying the right input at the right place at the right time in the right amount.

Site-Specific Weed Management (SSWM) applies control measures only where weeds are present. Research by Mortensen et al. (1990s) showed that weed populations are highly aggregated, occupying only 10–30% of field area, creating substantial opportunities for targeted management and chemical reduction.

First-generation SSWM used optical sensors detecting green plants against brown soil ('Green-on-Brown'). These failed in 'Green-on-Green' scenarios where weeds grow amongst crops, leading to shape-based classification using computer vision. However, traditional image processing with hand-crafted features lacked robustness for practical field deployment.

2.3 Deep Learning in Object Detection

Deep Learning, particularly Convolutional Neural Networks (CNNs), revolutionized computer vision by enabling automated feature learning from raw images. Unlike traditional machine learning requiring manual feature engineering, deep learning discovers optimal features through training on large labeled datasets. CNNs employ multiple layers of learned filters detecting progressively complex features: early layers identify edges and textures, intermediate layers recognize parts and patterns, and deeper layers capture complete objects and semantic concepts.

2.3.1 The YOLO Architecture

You Only Look Once (YOLO) by Redmon et al. (2016) reimagined object detection as a single regression problem. Unlike R-CNN's two-stage approach, YOLO predicts bounding boxes and class probabilities directly from full images in one forward pass. YOLO divides images into an S×S grid; each grid cell predicts multiple bounding boxes, confidence scores, and conditional class probabilities. Final detections use non-maximum suppression to eliminate duplicates. This unified architecture processes images at 45–155 frames per second, enabling real-time agricultural applications.

YOLO-NAS (Neural Architecture Search) by Deci AI, implemented via the SuperGradients framework, represents a further evolution in the YOLO family. Unlike previous versions that relied on manually designed backbones, YOLO-NAS applies automated Neural Architecture Search to discover an optimized network structure specifically engineered for superior accuracy-latency trade-offs. It introduces Quantization-Aware RepVGG (QA-RepVGG) blocks enabling efficient int8 quantization at inference without accuracy loss, and a hybrid attention mechanism improving detection of small and overlapping objects. YOLO-NAS achieves higher mAP scores than YOLOv8 on standard COCO benchmarks while maintaining real-time inference speed making it particularly suitable for agricultural weed detection where fine-grained discrimination between morphologically similar plant species is required.

2.4 IoT Hardware for Edge AI

2.4.1 ESP-32 Microcontroller Platform

Deploying AI in agricultural fields requires Edge Computing processing data locally rather than transmitting to cloud servers. Edge AI eliminates network latency, reduces bandwidth requirements, ensures operation without continuous internet connectivity, and addresses data privacy concerns.

The ESP-32 is a low-cost, low-power microcontroller by Espressif Systems featuring a dual-core Xtensa LX6 processor (240 MHz), integrated WiFi (802.11 b/g/n) and Bluetooth, and multiple peripheral interfaces (UART, SPI, I2C, GPIO). It is available in Kenya for approximately KES 1,400 with extensive open-source library support.

While the ESP-32 cannot run complex deep learning models due to its limited 520 KB RAM, it excels as a control interface. In this project, the ESP-32 receives detection results from the AI model running on a more powerful computer and actuates the LED. This distributed architecture leverages the ESP-32's strengths in low-power operation, peripheral control, and wireless connectivity while offloading AI inference to appropriate hardware.

CHAPTER 3: MATERIALS AND METHODS

3.1 Preliminary Investigations

Prior to system development, a comprehensive feasibility assessment was conducted across three key areas. Image data was collected directly from a castor bean farm in Kiambu County to characterize the weed species present, their density, and seasonal distribution patterns under real field conditions. A literature review on prior AI-based weed detection systems identified YOLO-NAS, implemented via the SuperGradients framework, as the most suitable model for real-time, resource-constrained deployment due to its superior accuracy-latency trade-off over prior YOLO architectures. Additionally, a hardware benchmarking study evaluated the ESP-32 microcontroller against alternatives such as Raspberry Pi Zero and Arduino Nano, confirming that the ESP-32 offered the optimal balance of cost, connectivity, and I/O capability for LED actuation tasks at a fraction of the cost of competing platforms.

3.2 Plant Material

The primary crop selected for this study is Castor Bean (Ricinus communis), a common cash crop in East African agriculture. Castor bean is characterized by its distinctive large, star-shaped palmate leaves with reddish-purple stems, making it visually distinct from most weed species. This crop was selected based on its prevalence in available agricultural datasets and its economic importance in the target farming regions.

The following weed species were identified based on their frequency and economic impact in castor bean cultivation farm:

#	Weed Species	Common Name	Frequency
1	Amaranthus viridis/hybridus	Pigweed / Green Amaranth	Very High
2	Silybum marianum	Milk Thistle	Low

Pigweed is classified as 'Very High' frequency due to its aggressive growth, rapid reproduction, and widespread presence across multiple agro-ecological zones. This species receives priority in the training dataset. The remaining species are included to ensure the model generalizes across diverse weed types and field conditions.

3.3 Hardware

The system hardware comprises two integrated layers:

Vision and Processing Subsystem: A digital camera with a minimum resolution of 12 megapixels for image capture, and a computer or cloud platform (Google Colab or a PC with GPU) running the trained

AI model. This layer performs the cognitive function of weed identification through real-time image analysis.

Actuation and Indication Subsystem: The ESP-32 microcontroller, current-limiting resistors, LED indicators, and jumper wires assembled on a breadboard. This layer receives detection signals and provides immediate visual feedback through LED activation.

Component	Specification	Quantity	Unit Cost (KES)	Total (KES)
ESP-32 Development Board	Dual-core, 240MHz,	1	1,400	1,400
Breadboard & Jumper Wires Set	830 tie-points, M-M wires	1	500	500
Resistors (220–330Ω)	1/4W, assorted set	1 Set	150	150
LEDs (5mm, Red/Green)	5mm standard	3	35	100
Google Colab Pro / GPU Access	3 months subscription	1	8,000	8,000
GRAND TOTAL				10,150

Circuit Connection Procedure:

1. Mount ESP-32 on the breadboard, spanning the center channel.
2. Connect the breadboard ground rail to the ESP-32 GND pin.
3. Insert LED with anode (longer leg) and cathode (shorter leg) in adjacent rows.
4. Connect a 220Ω resistor between the LED anode and GPIO pin
5. Connect the LED cathode to the ground rail.
6. Connect ESP-32 to computer via USB for power and programming.

3.4 Software

The software stack consists of the following components:

- Python 3.8+ with SuperGradients: The Deci AI SuperGradients framework serves as the core deep learning library for model training and inference. SuperGradients provides production-ready training recipes, built-in COCO dataloader utilities, and native YOLO-NAS model implementations.
- YOLO-NAS (via SuperGradients): The primary object detection model. Three model variants are evaluated — YOLO-NAS Small (yolo_nas_s), YOLO-NAS Medium (yolo_nas_m), and YOLO-NAS Large (yolo_nas_l) — pre-trained on the COCO dataset and fine-tuned on the custom weed/crop dataset.
- Google Colab: Cloud-based GPU environment for AI model training (NVIDIA T4/A100 acceleration).
- Roboflow: Web-based platform for dataset management, annotation, augmentation, and export in YOLO/COCO format. Used for preliminary segmentation investigations as detailed in Section 3.6.3.
- LabelImg: Desktop annotation tool for creating YOLO-format bounding box labels offline.

- Arduino IDE / MicroPython: Firmware development environment for ESP-32 microcontroller programming.
- OpenCV: Computer vision library for image capture, preprocessing, and frame extraction from video feed.

3.5 Algorithm Development

The complete detection pipeline is implemented as a closed-loop control system:

1. Image Capture: Camera captures video feed at 30 frames per second; each frame is extracted as an independent image.
2. Preprocessing: Frames are resized to 640×640 pixels (optimal input dimension for YOLO-NAS), pixel values normalized from range 0–255 to 0–1, and color space maintained as RGB.
3. AI Inference: Preprocessed frames are fed into the trained YOLO-NAS model. Forward propagation through the QA-RepVGG blocks and hybrid attention layers generates predictions comprising bounding box coordinates, class label, and confidence score (0–1) for each detected object.
4. Confidence Thresholding: Only detections with a confidence score above 85% are accepted as valid weed detections, minimizing false positives.
5. Signal Generation: If at least one weed detection exceeds the threshold, the AI generates a digital HIGH (TRUE) signal transmitted to the ESP-32 via UART or WiFi.
6. LED Actuation: Upon receiving the TRUE signal, the ESP-32 sets the GPIO pin HIGH for 2,000 ms (2 seconds), illuminating the LED before returning to the monitoring state.

3.6 Main Image Processing

3.6.1 Color Space Transformation

While RGB is the native image format for the camera, colour space transformation is applied at specific processing stages to enhance feature discrimination:

- RGB to HSV (Hue, Saturation, Value): Applied during preprocessing to isolate vegetation using the Excess Green Index. HSV decouples colour information from intensity, making the system more robust to varying lighting conditions in the field.
- RGB to LAB: Employed during data augmentation to apply perceptually uniform colour jitter, improving model generalization across different soil types and lighting environments.
- Greyscale Conversion: Applied in contour-based preprocessing to enable edge detection for initial plant boundary identification independent of colour information.

The primary inference pipeline retains the RGB colour space, as YOLO-NAS is trained and optimized on RGB inputs. Transformation layers are applied as preprocessing augmentation steps to expand the effective training dataset distribution.

3.6.2 Preprocessing

The preprocessing pipeline standardizes all input images before model inference:

- Resizing: All images are resized to 640×640 pixels, the optimal input dimension for YOLO-NAS using bilinear interpolation to maintain aspect ratio where possible.

- Normalization: Pixel intensity values are normalized from the range [0, 255] to [0.0, 1.0] by dividing by 255, ensuring consistent gradient magnitudes during inference.
- Colour Normalization: Mean subtraction using ImageNet statistics is applied for transfer learning compatibility.
- Noise Reduction: A Gaussian blur filter (kernel size 3×3) is applied to raw field images to reduce sensor noise prior to annotation and training.

3.6.3 Segmentation and Dataset Annotation

Image segmentation is the process of partitioning each image into meaningful regions corresponding to individual plant instances. For this project, instance segmentation masks were generated using Roboflow's annotation platform as part of the preliminary dataset investigation.

A sample dataset of bean plant images was annotated and exported in COCO segmentation format via Roboflow. The dataset comprised 4 images with the following segmentation metadata:

Image ID	File Name	Dimensions	Annotated Class	Segmentation Type
0	agri_01.jpeg	512 × 512 px	bean_plant	Instance (RLE mask)
1	agri_02.jpeg	512 × 512 px	bean_plant	Instance (RLE mask)
2	agri_03.jpeg	512 × 512 px	bean_plant	Instance (RLE mask)
3	agri_04.jpeg	512 × 512 px	bean_plant	Instance (RLE mask)

Each image contains one annotated plant instance with a polygon segmentation mask encoded in Run-Length Encoding (RLE) format. The segmentation masks define precise plant boundaries independent of rectangular bounding boxes, enabling more accurate shape-based feature learning. The COCO JSON structure used includes image metadata, category definitions (bean_plant_1.0 as supercategory; bean_plant as the target class), and per-annotation RLE segmentation counts with associated area values.

For the main YOLO-NAS training pipeline, the annotation format transitions from COCO segmentation to YOLO bounding box format using the standard YOLO annotation structure:

<class_id> <x_center> <y_center> <width> <height>

Where class_id is an integer (0 = Castor Bean, 1 = Amaranthus viridis/hybridus) and all coordinate values are normalized between 0.0 and 1.0 by dividing pixel coordinates by the image dimensions, ensuring annotation files are resolution-independent.

The primary annotation tool selected is LabelImg due to its widespread adoption, direct YOLO format export, and offline desktop functionality suitable for areas with limited internet connectivity. Additional tools evaluated include:

Tool	Free	Format	Notes
LabelImg	Yes	YOLO	Most popular; desktop app; selected for this project
Roboflow	Freemium	YOLO/COCO	Web-based; easy export; used for preliminary segmentation
CVAT	Yes	YOLO/COCO	Feature-rich; web-based
Label Studio	Yes	Multiple	Very flexible; API support

Dataset Organization: The annotated dataset will be organized in the standard YOLO directory structure with images/ and labels/ subdirectories partitioned into train/ (70%), val/ (20%), and test/ (10%) splits.

3.6.4 Feature Extraction

Feature extraction is performed automatically by the YOLO-NAS backbone through hierarchical convolutional processing via Quantization-Aware blocks:

- Low-level features (Layers 1–4): Edge detection, colour gradients, and basic texture primitives such as leaf surface patterns and venation.
- Mid-level features (Layers 5–10): Plant organ shapes (leaf lobes, stem cross-sections), species-specific texture patterns, and structural arrangements.
- High-level features (Layers 11+): Complete plant morphology, species-discriminative semantic representations, and contextual spatial relationships between plant instances.

Transfer learning from COCO-pretrained YOLO-NAS weights initializes the feature extraction backbone with general visual features. Fine-tuning on the custom crop/weed dataset adapts these features to the specific morphological characteristics of Kenyan agricultural species.

Key morphological features exploited for weed-crop discrimination include leaf shape index (roundness, elongation, lobing), leaf texture features (surface roughness, venation density), colour histograms in HSV space (hue distribution for Amaranthus' reddish tones versus castor bean's deep green), and plant architectural features (branching patterns, stem diameter relative to height).

CHAPTER 4: EXPECTED RESULTS

4.1 High-Precision AI Detection Model

The trained YOLO-NAS model is expected to identify weeds versus crops with a minimum mean Average Precision (mAP) of 85% on the held-out test dataset.

The 85% mAP threshold represents the minimum accuracy level required for practical precision farming deployment. At this performance level, the system approaches human expert accuracy while operating at speeds impossible for manual inspection.

4.2 Responsive LED Indicator System

The IoT LED actuation system is expected to demonstrate:

- LED activation within 50 ms of signal reception from the AI processor (near-instantaneous feedback).
- Precisely controlled 2-second blink duration per detection event.
- LED activation only when weed detection confidence exceeds 85%, avoiding false activations.
- Automatic OFF state after 2 seconds, immediately ready for subsequent signals without manual reset.
- Total system latency under 500 ms, enabling real-time agricultural response.
- Reliable operation for extended periods without performance degradation.

4.3 Integrated Prototype Validation

The fully assembled system will be validated through systematic testing with the following expected performance outcomes:

Performance Metric	Target	Measurement Method
True Positive Rate	≥ 80%	LED activates correctly when weeds present
False Positive Rate	< 15%	LED does not activate without weeds present
System Uptime	> 95%	Operational time without failures over 4-hour session
Response Latency	< 500 ms	Time from image capture to LED activation
LED Timing Accuracy	2,000 ms ± 10 ms	Oscilloscope-measured blink duration

4.4 Project Implications and Contribution to Knowledge

Economic Impact: Upon future mechanical spraying integration, the validated system is projected to deliver herbicide cost savings of 60–70% (KES 5,000–10,000/hectare/season), labor cost reductions through automated detection (KES 15,000–30,000/hectare/season), and yield protection by preventing 30–40% crop losses (KES 30,000–60,000/hectare revenue).

Environmental Sustainability: Transition from blanket spraying to site-specific application will reduce soil and water chemical contamination, protect beneficial soil microorganisms and pollinators, and slow the evolution of herbicide-resistant weed populations.

Academic Contribution: The project will produce the first comprehensive annotated dataset of Kenyan crops and weeds adapted to East African conditions, establish baseline performance metrics for AI-based weed detection in Kenyan maize systems, and provide a replicable methodology for AI-IoT integration in resource-constrained agricultural contexts.

MATERIALS AND BUDGET

Project Budget

The following table outlines the estimated costs for all components and services required for successful project completion. All costs are in Kenya Shillings (KES) based on current market prices from local electronics suppliers in Nairobi as of January 2026.

Item Description	Quantity	Unit Cost (KES)	Total Cost (KES)	Justification
Breadboard & Jumper Wires Set	1	500	500	Tool-free prototyping platform
ESP-32 Development Board	1	1,400	1,400	Core IoT microcontroller for LED actuation
Resistors (220–330Ω, 1/4W assorted)	1 Set	150	150	Current limiting for LED circuit
LEDs (5mm, Red/Green)	3	35	100	Visual weed detection indicators
Google Colab Pro / GPU Access (3 months)	1	8,000	8,000	NVIDIA T4/A100 GPU for AI model training
GRAND TOTAL			10,150	

Budget Notes and Justification:

- ESP-32 was selected over Raspberry Pi (KES 8,000+) due to sufficient GPIO functionality at one-sixth the cost for the actuation control task.
- Google Colab Pro subscription provides GPU resources (NVIDIA T4/A100) for AI training at a fraction of dedicated GPU hardware cost (KES 80,000–200,000).
- Breadboard-based prototyping allows rapid iteration without custom PCB fabrication, reducing cost and development time.
- Budget excludes camera equipment as the project utilizes existing smartphones/DSLR cameras available to the research team.
- Transportation for field data collection is absorbed as institutional research support.

Project Work Plan

The project is planned for completion within 20 weeks (January 2026 to May 2026), structured into 8 sequential phases:

Smart Weed Detection Project - Gantt Chart

Liceria & Co. | Team: Amos Kibet & Paul Kimanga M | January - May 2026

Task ID	Task Name	Start Date	End Date	Duration (Days)
1	Research & Literature Review	2025-12-20	2026-01-20	31
2	Data Collection, EDA & Cleaning	2026-01-21	2026-02-10	20
3	Modeling	2026-02-11	2026-03-03	20
4	Evaluation of the Model	2026-03-04	2026-03-17	13
5	ESP32 & LED Integration	2026-03-18	2026-04-07	20
6	Testing & Evaluation	2026-04-08	2026-04-21	13
7	Documentation	2026-04-22	2026-05-05	13
8	Presentation & Submission	2026-05-06	2026-05-20	14

Phase	Activity	Duration	Deliverable
1	Research & Literature Review	Weeks 1–4	Literature review report
2	Data Collection, EDA & Cleaning	Weeks 5–7	Annotated dataset
3	Model Training	Weeks 8–10	Trained YOLO-NAS weights (ckpt_best.pth)
4	Model Evaluation	Weeks 11–12	Model evaluation report with metrics
5	ESP-32 & LED Integration	Weeks 13–16	Functioning LED indicator system
6	System Testing & Evaluation	Weeks 17–18	System performance validation report
7	Documentation	Weeks 19–20	Complete project report
8	Presentation & Submission	Weeks 21–22	Defended project and submitted report

REFERENCES

1. Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). You Only Look Once: Unified, Real-Time Object Detection. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) (pp. 779–788). IEEE. https://doi.org/10.1109/CVPR.2016.91

2. Deci AI. (2023). SuperGradients: A Production-Ready Deep Learning Training Library. GitHub. Retrieved from https://github.com/Deci-AI/super-gradients

3. Partel, V., Charan Kakarla, S., & Ampatzidis, Y. (2019). Development and evaluation of a low-cost and smart technology for precision weed management utilizing artificial intelligence. Computers and Electronics in Agriculture, 157, 339–350.

4. Islam, N., et al. (2021). Early Weed Detection Using Image Processing and Machine Learning Techniques in an Australian Chilli Farm. Agriculture, 11(5), 387.

5. Hasan, A. S. M., Sohel, F., Diepeveen, D., Laga, H., & Jones, M. G. (2021). A survey of deep learning techniques for weed detection from images. Computers and Electronics in Agriculture, 184, 106067.

6. Wang, A., Xu, Y., Wei, X., & Cui, B. (2020). Semantic Segmentation of Crop and Weed using an Encoder-Decoder Network and Image Enhancement Method under Uncontrolled Outdoor Illumination. IEEE Access, 8, 81735–81746.

7. Oberti, R. & Schillaci, G. (2020). Precision Agriculture: Sensing, Data Analysis and Actuation. In Agricultural Internet of Things and Decision Support for Precision Smart Farming (pp. 1–30). Academic Press.

8. Espressif Systems. (2024). ESP32 Series Datasheet: 2.4 GHz Wi-Fi + Bluetooth + Bluetooth LE SoC. Retrieved from https://www.espressif.com

9. Dhanaraju, M., Chenniappan, P., Ramalingam, K., Pazhanivelan, S., & Kaliaperumal, R. (2022). Smart Farming: Internet of Things (IoT) and Artificial Intelligence (AI) in Agriculture. Agronomy, 12(11), 2713.

10. Gao, J., Nuyttens, D., Lootens, P., He, Y., & Pieters, J. G. (2018). Recognising weeds in a maize crop using a random forest machine-learning algorithm and an overhead video sensor. Biosystems Engineering, 166, 38–52.

11. Lottes, P., Behley, J., Milioto, A., & Stachniss, C. (2018). Fully convolutional networks with sequential information for robust crop-weed detection in precision farming. IEEE Robotics and Automation Letters, 3(4), 2870–2877.

12. Teimouri, N., et al. (2018). Weed growth stage estimator using deep convolutional neural networks. Sensors, 18(5), 1580.

13. Kamilaris, A., & Prenafeta-Boldú, F. X. (2018). Deep learning in agriculture: A survey. Computers and Electronics in Agriculture, 147, 70–90.

14. Terven, J., & Cordova-Esparza, D. (2023). A Comprehensive Review of YOLO Architectures in Computer Vision: From YOLOv1 to YOLOv8 and YOLO-NAS. Machine Learning and Knowledge Extraction, 5(4), 1680–1716. 15. Aharon, S., Louis-Dupont, Ofri-Mann, et al. (2021). SG: Deci SuperGradients. GitHub. https://github.com/Deci-AI/super-gradients. YOLO-NAS: A Foundation Model for Object Detection. Deci AI. (2023). Retrieved from https://deci.ai/blog/yolo-nas-object-detection-foundation-model/

15. Bochkovskiy, A., Wang, C. Y., & Liao, H. Y. M. (2020). YOLOv4: Optimal Speed and Accuracy of Object Detection. arXiv preprint arXiv:2004.10934.

16. Milioto, A., Lottes, P., & Stachniss, C. (2018). Real-time semantic segmentation of crop and weed for precision agriculture robots leveraging background knowledge in CNNs. In 2018 IEEE International Conference on Robotics and Automation (ICRA) (pp. 2229–2235). IEEE.